

1. Requirements Documentation

1.1 Product Vision and Product Goals:

Product Vision

This project is a fun, safe, and educational memory matching game designed for children aged 3 to 7. The main goal is to help young users develop their memory, attention, visual recognition, and matching skills, all while enjoying a visually appealing experience. With lovingly designed animal-themed cards, the game offers both entertainment and learning opportunities.

Product Goals

- Provide scalable difficulty levels (from 8 to 24 pairs) suitable for children aged 3–7
- Include custom-designed animal-themed cards, all in a consistent art style
- Support both touch and mouse inputs for accessibility across various devices
- Stimulate memory, focus, and visual association skills
- Offer a simple, intuitive interface tailored for early childhood motor skills
- Ensure seamless playability across tablets, smartphones, and computers

1.2 Personas

Lily (4)

- Attends kindergarten
- Loves animals and colorful visuals
- Has not yet developed fine motor skills for using a mouse
- Enjoys playing simple games on her family's tablet
- Her parents seek high-quality, educational digital content

Jack (6)

- Recently started elementary school
- Can comfortably use both touchscreens and a mouse
- Enjoys memory games and likes to challenge himself with harder levels
- Loves learning new animals through interactive experiences

Emma (parent)

- Wants to find age-appropriate, safe, ad-free digital games
- Values educational and developmentally supportive content
- Prefers games that don't overstimulate but provide meaningful engagement

1.3 User Stories

- As a player who cannot use a mouse yet, I want to flip cards just by tapping the screen.
- As a player, I want to see cute animals on the cards so that the game feels fun and exciting.
- As a player, I want to choose the difficulty level so I can challenge myself and improve.
- As a player, I want the cards to be different each time I play so that it stays interesting.
- As a parent, I want the game to be safe and appropriate for my child's age so I can let them play with peace of mind.
- As a parent, I want the game to support my child's cognitive development so that screen time is beneficial.
- As a player, I want audio feedback and a calming theme music so I can engage with the game without being distracted.
- As a parent, I want the game to be ad-free without any payment so my child won't get distracted by other app offers/inappropriate content.
- As a player, I want to receive a score at the end of each level depending on my clicks and time left so I can get motivated to get better results.

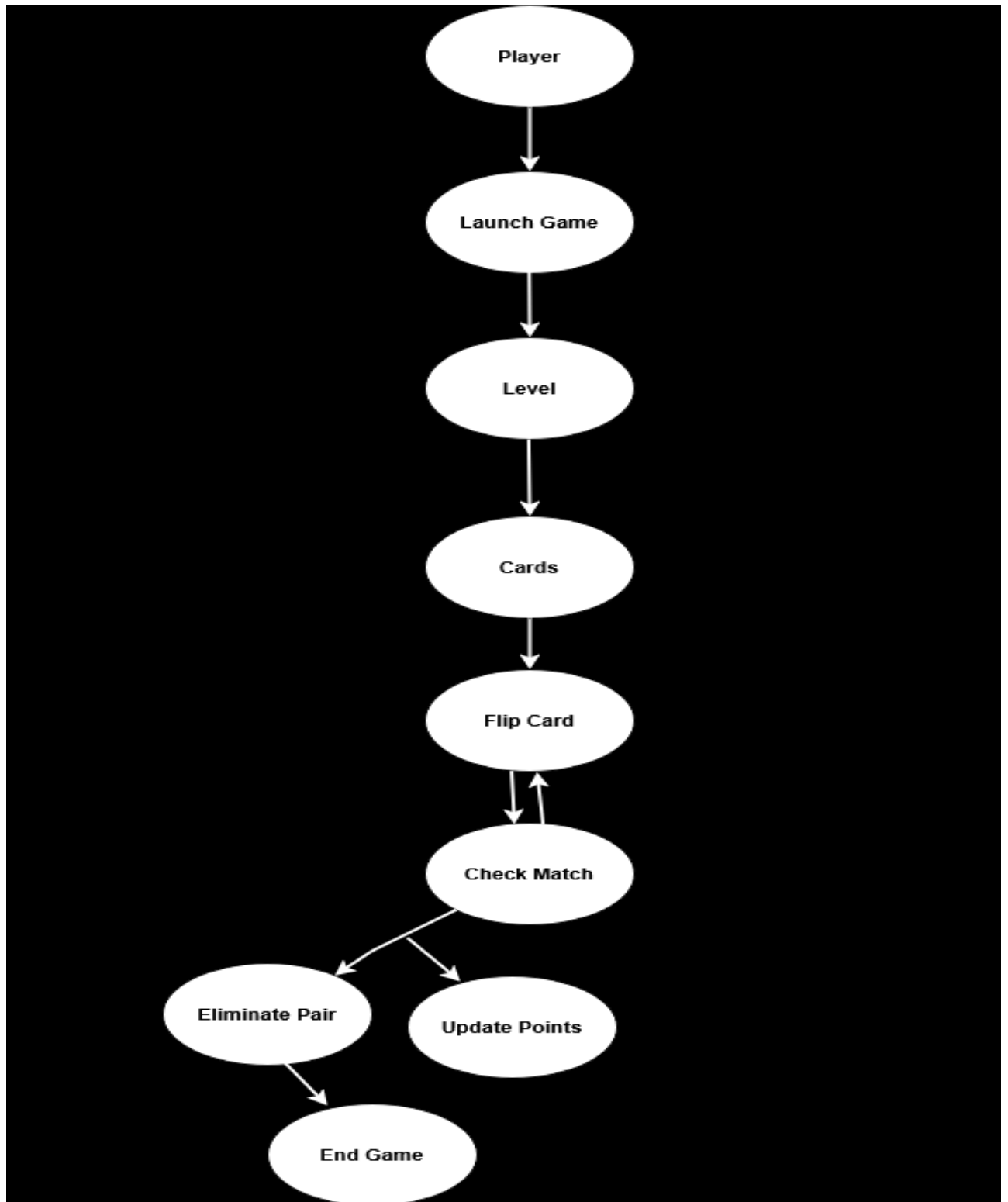
1.4 Use Case Descriptions & Diagram

Use Cases Descriptions : The table below provides a description of each use case in the game :

Use Case	Actor	Precondition	Main Flow	Postcondition
Launch game	Player	none	Player runs the application on a device	Main Menu is displayed
Level selection	Player	Main Menu is displayed	Player selects a level from 3 options	Level launches
Level launch	System	Input has been taken to launch a level	Cards and HUD elements are displayed	Waiting for player input
Flip Card	Player	Level has been launched	Player clicks on random card	Card is flipped and shown to the player.
Track Time and clicks of players	System	players have flipped the first card.	Timer starts, ClicksRegister starts	clicks are updated,Timer is running.

Match cards	System	Two cards are flipped	System checks if both flipped cards match	Both cards get eliminated if same if not cards are flipped again
Update score	System	Game must be in progress	counts the score as per scoring algorithm	The score is updated at every card match according to the scoring algorithm.
Register that all cards have been paired	System	All cards have been eliminated	Display final score and message	Waiting for players input to redirect to main menu

Use Case Diagram(Text-Based):



1.5 Quantity structure

Element	Details
Total Difficulty Levels	3 levels (starting from 8 pairs to 24 pairs)
Total Card Designs	24 unique animal cards
Cards per Game (Varies by level)	16 cards (8 pairs) to 48 cards (24 pairs)
Input Methods	2 types (mouse and touch Input)
Supported Platforms	2 types (desktop and Android)
Languages	1 (English, with potential for future localization)
Audio Elements	2 sound effects and 1 theme music

1.6 Dictionary/Glossary of terms

Term	Definition
Pair Matching Game	A memory game where the player flips cards to find matching pairs.
Card	A visual object in the game featuring a hidden image (animal-themed) that is revealed when flipped.
Pair	Two cards with identical animal images. The objective is to find all matching pairs
Difficulty Level	A setting that determines the number of card pairs used in a game round. Higher levels have more pairs, increasing complexity.
Animal Theme	The visual style and subject matter of the cards, which depict animals in a consistent, custom-designed art style.
Touch Input	User input via screen tap, used on mobile and tablet devices.
Mouse Input	User input via click, used primarily on desktop devices.
Game Session	A single playthrough of the game, from start to either successful completion or early exit.
UI (User Interface)	The visual elements and buttons that the player interacts with (e.g, level selection, main menu, mute/unmute)
Pointing System	A pointing system based on the total amount of clicks and time used per level.